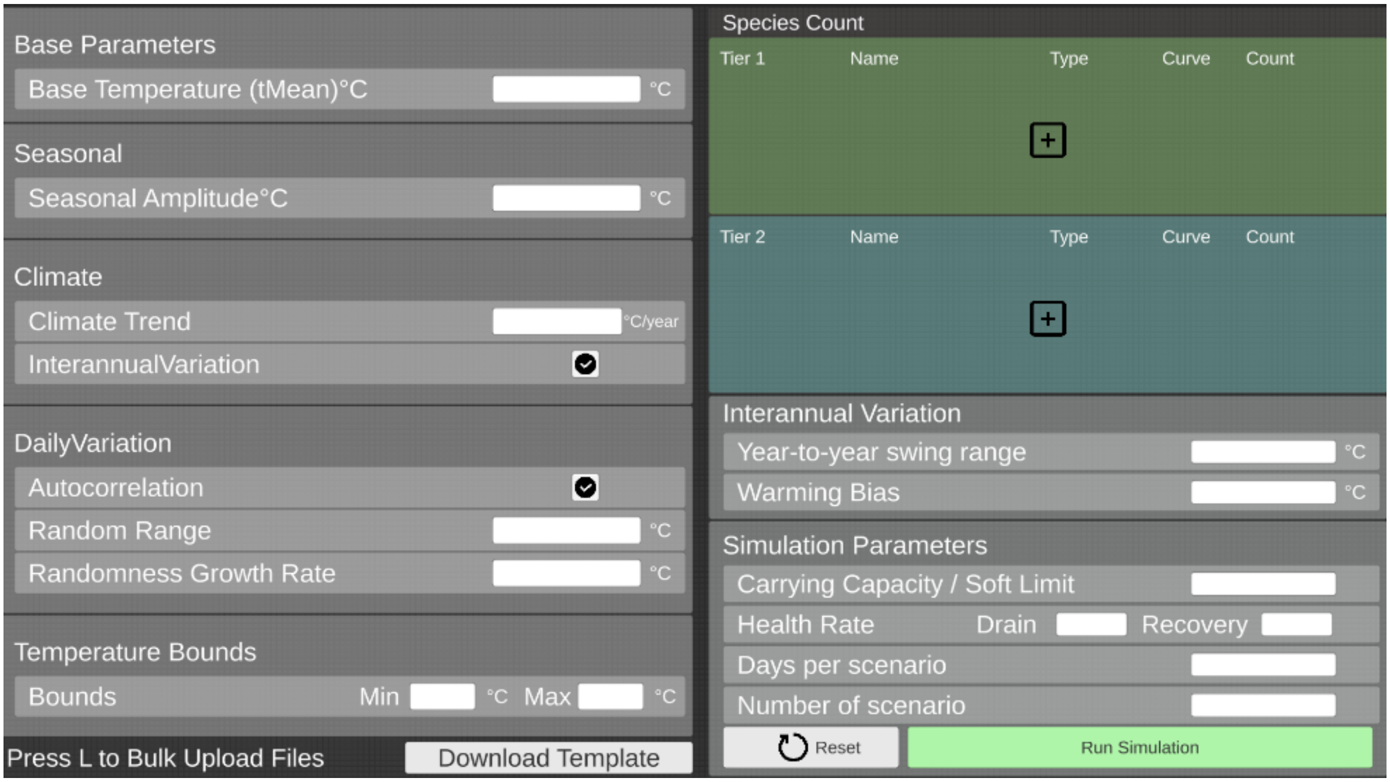
TinySea Simulation Guide

Reference for UI fields, CSV formats, and output data



Simulation setup screen — left panel (temperature) and right panel (species & parameters)

1. Simulation Setup (Left Panel)

These fields appear on the left side of the simulation configuration screen. All temperatures are in Celsius.

Base Parameters

UI Field	Default	Description
Base Temperature (tMean)	20	The average/mean annual temperature. The simulation oscillates around this value seasonally.

Seasonal

UI Field	Default	Description
Seasonal Amplitude	10	How much temperature swings above/below the base each year. 10 means the range is roughly base +/- 10 degrees across summer and winter.

Climate

UI Field	Default	Description
Climate Trend	1	Degrees of warming per year (linear). 0 = stable climate. 0.02 = slow warming. Negative = cooling.
Interannual Variation	On	When checked, each simulated year gets a random temperature offset, making some years warmer/cooler than others.

Daily Variation

UI Field	Default	Description
Autocorrelation	On	When checked, day-to-day temperature changes are smooth (like real weather). When off, each day is fully random.
Random Range	5	Base daily temperature variation range in degrees. Higher = more day-to-day noise.
Randomness Growth Rate	0.5	How much the daily random range increases each year. Models increasing climate instability over time.

Temperature Bounds

UI Field	Default	Description
Min	-5	Hard floor -- temperature never drops below this value.
Max	50	Hard ceiling -- temperature never exceeds this value.

1b. Simulation Setup (Right Panel)

Species Count

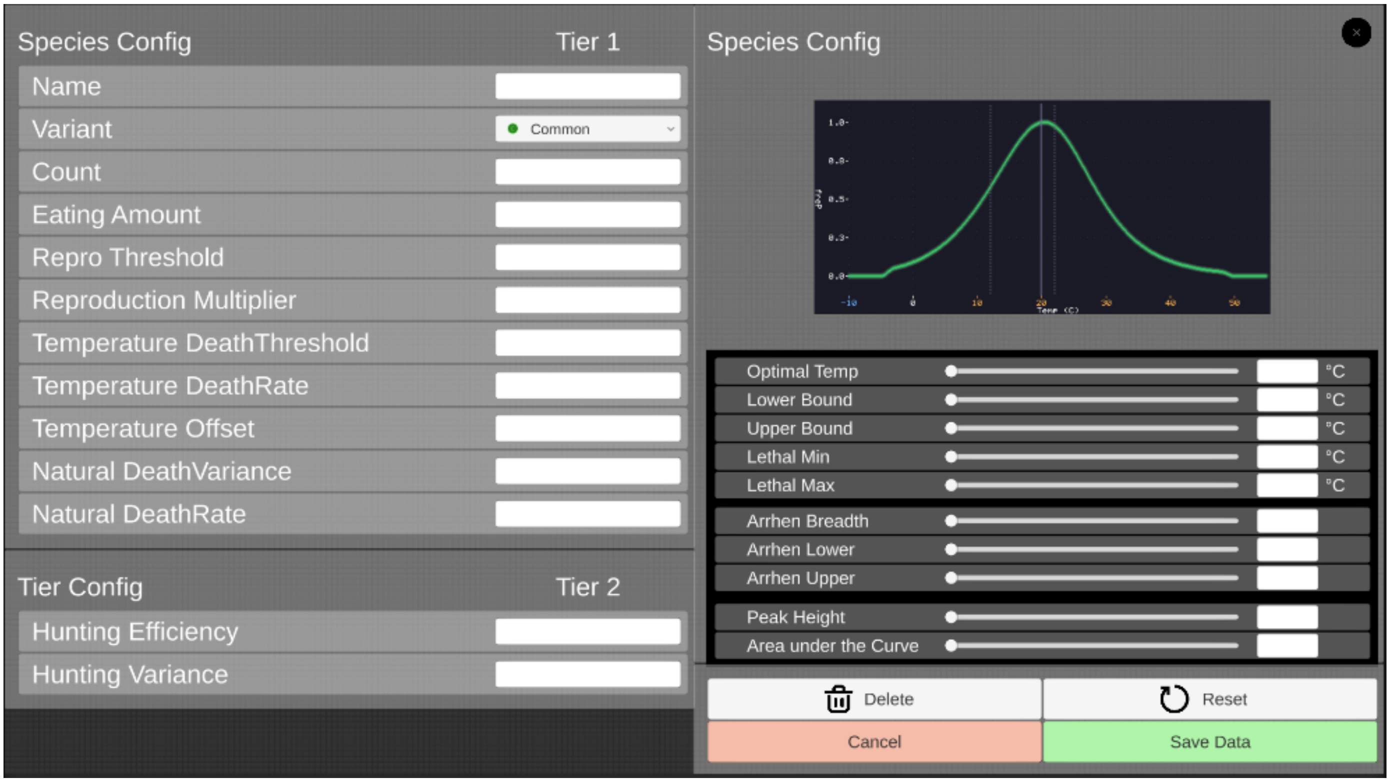
The Tier 1 and Tier 2 tables show species in the simulation. Click the [+] button to add a species. Click a species row to edit it (opens the Species Config screen). Tier 1 = prey, Tier 2 = predators. Each species has a Name, Variant type (Arctic/Common/Tropical/Custom), a thermal curve preview, and starting population Count.

Interannual Variation

UI Field	Default	Description
Year-to-year swing range	2	Magnitude of year-to-year random temperature offset. Higher = more variation between years.
Warming Bias	1.5	Skews the year-to-year variation toward warmer years. 0 = symmetric. Higher = warmer years more likely.

Simulation Parameters

UI Field	Default	Description
Carrying Capacity / Soft Limit	5000	Maximum sustainable Tier 1 population. As prey approaches this number, their birth rate decreases. Does NOT kill creatures -- just slows reproduction.
Health Rate: Drain	0.15	How fast species health declines when temperature is bad. Higher = faster decline.
Health Rate: Recovery	0.10	How fast species health recovers when temperature is good. Intentionally slower than drain (harder to recover than to get sick).
Days per scenario	365	How many days each simulation scenario runs. 365 = 1 year.
Number of scenarios	5	How many times to run the simulation with different random seeds. More scenarios = better statistics.



Species configuration — biology parameters (left) and thermal curve editor (right)

2. Species Configuration

This screen opens when you click a species in the Species Count table, or click [+] to add one. Left side has biology parameters, right side has the thermal curve editor with a live preview graph.

Biology Parameters (Left)

UI Field	Prey Default	Predator Default	Description
Name	--	--	Species name (e.g. Hexapod, Sheplik, or any custom name).
Variant	Common	Common	Thermal type: Arctic (cold-adapted), Common (generalist), Tropical (warm-adapted), or Custom.
Count	1000	100	Starting population at the beginning of the simulation.
Eating Amount	0	1.5	Prey consumed per creature per day. Always 0 for prey (Tier 1).
Repro Threshold	0.25	0.25	Health level where reproduction transitions from struggling to healthy. Not a hard gate -- reproduction can still occur below this, just at very low rates.
Reproduction Multiplier	0.45	0.1	Birth rate multiplier. Higher = faster reproduction. Predators reproduce ~4.5x slower than prey.
Temperature DeathThreshold	0.3	0.3	Health level below which species start dying from thermal stress. Above this = safe.
Temperature DeathRate	0.6	0.3	Max fraction that die per day when health is below the death threshold. Prey die faster (less physiological buffering).
Temperature Offset	0	0	Shifts the temperature this species "feels." Positive = feels warmer than ambient.
Natural DeathVariance	0.01	0.005	Random daily fluctuation in natural death rate.
Natural DeathRate	0.02	0.01	Base death rate from old age/disease (always active regardless of temperature). 0.02 = 2% per day.

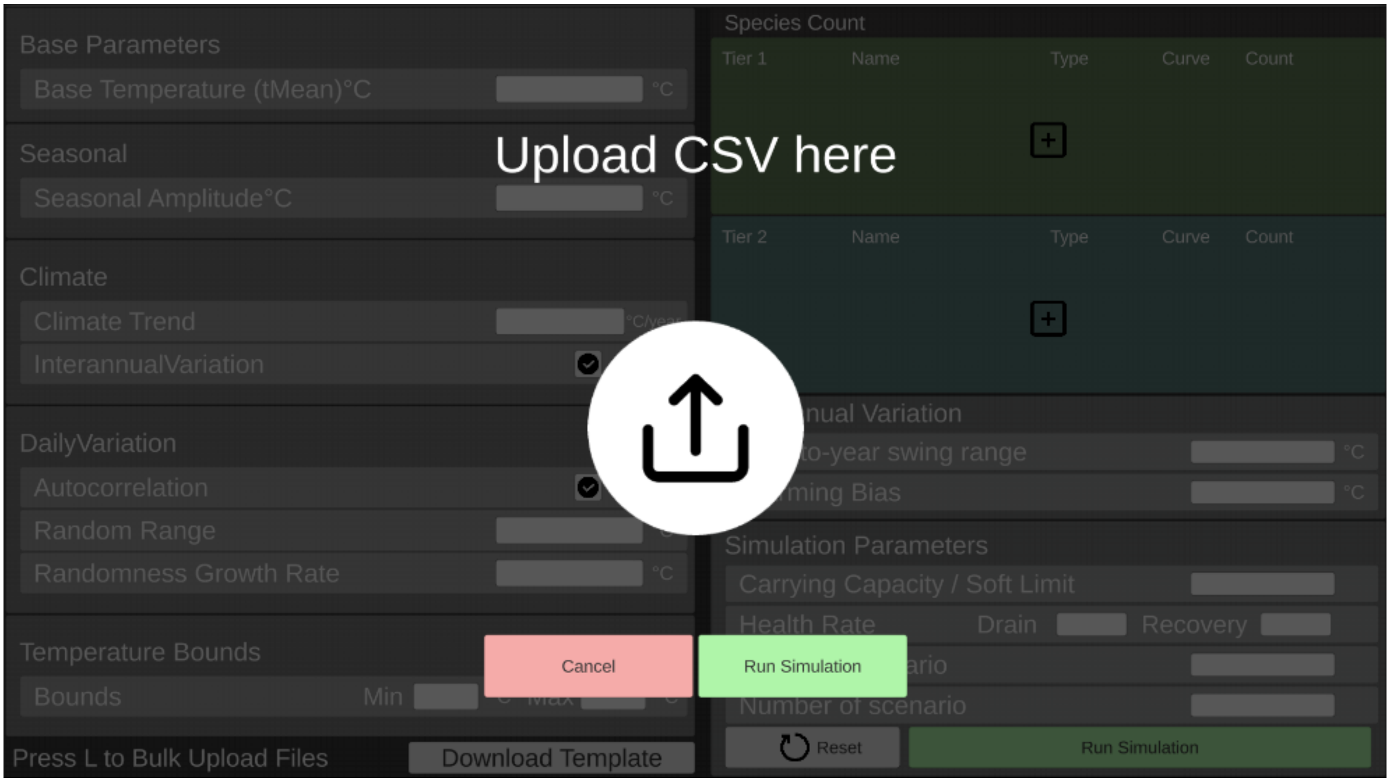
Tier 2 Only (Predator)

UI Field	Default	Description
Hunting Efficiency	0.75	Base hunting success rate (75%). Adjusted by prey availability -- more prey = easier hunting, fewer prey = harder.
Hunting Variance	0.15	Daily random fluctuation in hunting success.

Thermal Curve Editor (Right)

The graph shows how well the species performs at each temperature. The green curve is the thermal performance curve -- the peak is where the species does best. Performance drops on both sides.

UI Field	Description
Optimal Temp	Temperature (in Celsius) where the species performs best. The peak of the curve.
Lower Bound	Cold-side inflection point. Below this, performance drops faster.
Upper Bound	Warm-side inflection point. Above this, performance drops faster.
Lethal Min	Below this temperature, the species dies immediately. A 2-degree fade zone applies near this limit.
Lethal Max	Above this temperature, the species dies immediately. Same 2-degree fade zone.
Arrhen Breadth	Width of the thermal curve. Higher = wider tolerance range. Common (8000) is broader than Arctic/Tropical (4000).
Arrhen Lower	Controls how steeply performance drops on the cold side.
Arrhen Upper	Controls how steeply performance drops on the warm side.
Peak Height (Pmax)	Maximum performance at optimal temperature. 0.65 for Common (generalist trade-off), 0.85 for Arctic/Tropical (specialist advantage).
Area under the Curve	Visual indicator only -- shows the total area under the performance curve.



Bulk CSV upload screen — drag-and-drop or press L to load a file

3. Bulk CSV Upload

Press **L** or drag-and-drop a CSV file onto the upload screen. Each row in the CSV defines one "run" with its own temperature settings and species. The simulation executes all rows and packages results as a ZIP download. Use **Download Template** to get a blank CSV with all columns.

Environment Columns (one value per row)

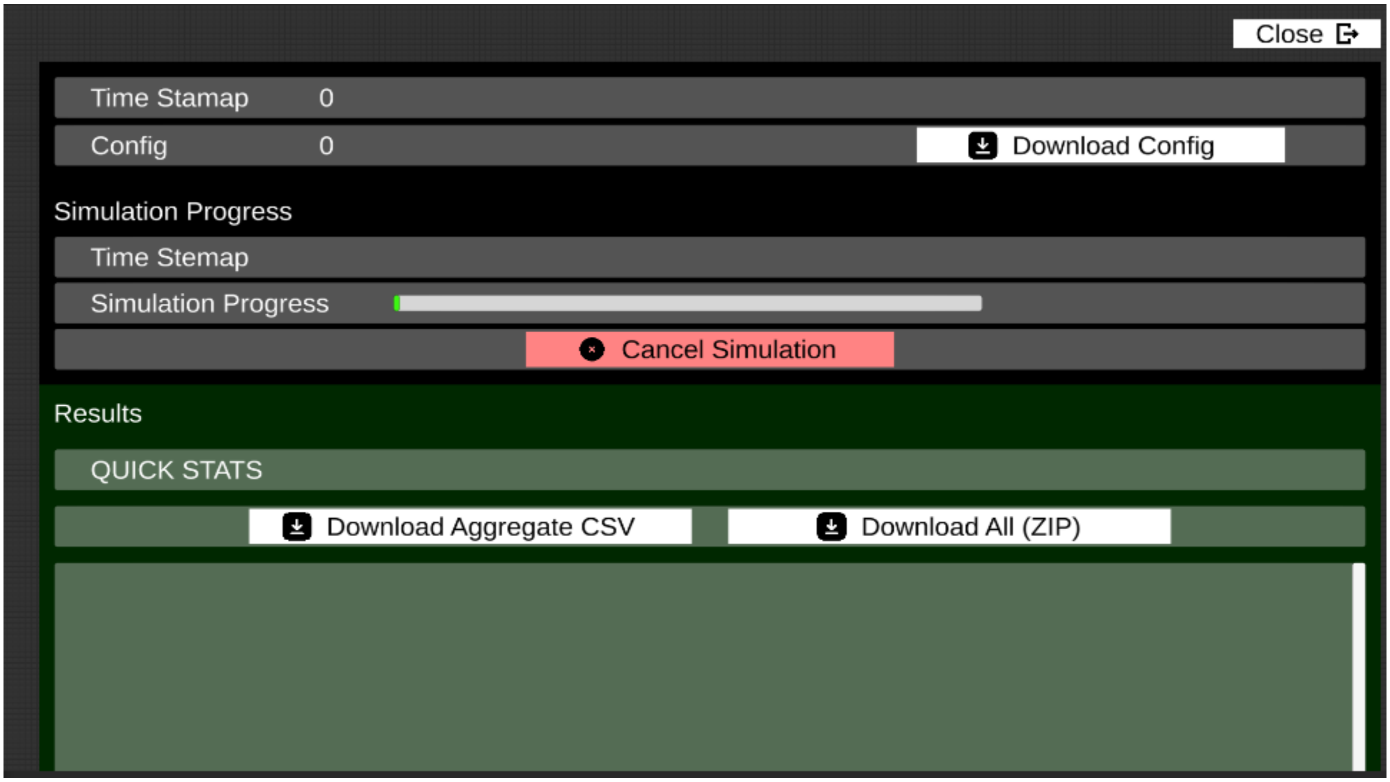
CSV Column	Description
batch_name	Label for this run (shows up in output filenames and summary).
days	Days per scenario (e.g. 365 for 1 year, 730 for 2 years).
num_scenarios	Number of scenarios per run (1-100). More = better statistics.
base_temp	Base temperature in Celsius.
seasonal_amp	Seasonal amplitude in Celsius.
climate_trend	Warming per year in Celsius (0 = stable, negative = cooling).
variability_mag	Year-to-year variation magnitude.
warming_bias	Bias toward warmer years.
daily_var_range	Daily random variation range.
randomness_growth	How much daily randomness increases per year.
autocorrelated	true/false -- smooth daily transitions.
interannual_variation	true/false -- enable year-to-year variation.
temp_min	Temperature floor (hard minimum).
temp_max	Temperature ceiling (hard maximum).
use_carrying_cap	true/false -- enable prey population soft limit.

CSV Column	Description
carrying_cap_t1	Tier 1 carrying capacity (required if carrying cap is on).
condition_drain_rate	Health drain rate (optional, default 0.15).
condition_recovery_rate	Health recovery rate (optional, default 0.10).

Species Columns (prefixed sp1_, sp2_, sp3_, ...)

Each species gets 20 columns with a numbered prefix. For example, the first species uses sp1_name, sp1_variant, sp1_tier, sp1_pop, etc. The second uses sp2_name, sp2_variant, and so on. Leave sp*_name blank to skip a slot. All temperatures in Celsius.

Column Suffix	Description
name	Species name (e.g. "Hexapod", "Coral", or any custom name).
variant	Arctic, Common, Tropical, or Custom.
tier	0 = Tier 1 (prey), 1 = Tier 2 (predator).
pop	Starting population.
eating	Prey consumed per creature per day (0 for prey).
repro_mult	Reproduction multiplier.
death_thresh	Health threshold for thermal death.
death_rate	Max death rate when below threshold.
repro_thresh	Health inflection for reproduction.
natural_death_rate	Base natural death rate.
natural_death_var	Natural death variance.
hunt_eff	Hunting efficiency (predators only).
hunt_var	Hunting variance.
opt_temp_c	Optimal temperature (Celsius).
arrhen_breadth	Arrhenius breadth (curve width).
arrhen_lower	Arrhenius lower (cold drop-off).
arrhen_upper	Arrhenius upper (warm drop-off).
lower_bound_c	Lower thermal bound (Celsius).
upper_bound_c	Upper thermal bound (Celsius).
pmax	Peak performance height (0-1). Optional, defaults to 1.0.



Results screen — progress, statistics, and download buttons

4. Understanding Results

After simulation completes, you see the Results screen with download buttons. There are two download options:

Download Options

Button	What You Get
Download Aggregate CSV	A single CSV with summary statistics across all scenarios for the current run.
Download All (ZIP)	A ZIP file with everything: individual scenario CSVs, aggregate CSV, config CSV, and (for bulk) a bulk_summary.csv at the root.

Terminology

Term	Definition
Scenario	One simulation execution with one random seed. The atomic unit. If you set "Number of scenarios" to 5, you get 5 scenarios.
Run	One configuration that generates N scenarios. In the UI: the current setup. In bulk mode: one CSV row.
Bulk	A collection of runs uploaded as a single CSV file. Each row = one run.

4a. Scenario CSV (one per scenario)

Each scenario file has configuration comments at the top (lines starting with #), then daily population data. R reads these cleanly with read.csv() since # is the default comment character.

Key columns in the daily data:

Column	Description
Day	Simulation day (1, 2, 3, ...).
Temperature	Temperature on this day (Celsius).
Tier1Pop / Tier2Pop	Total prey / predator population at end of day.
Tier1Arctic / Tier1Common / Tier1Tropical / Tier1Custom	Population breakdown by variant for Tier 1.
Tier2Arctic / Tier2Common / Tier2Tropical / Tier2Custom	Same breakdown for Tier 2. Tier*Custom = sum of all custom species in that tier.
EatenT1	Prey eaten by predators this day.
TempDeathsT1 / TempDeathsT2	Deaths from lethal temperatures (instant kill at extreme temps).
ConditionDeathsT1 / ConditionDeathsT2	Deaths from chronic thermal stress (health drops below death threshold).
NaturalDeathsT1 / NaturalDeathsT2	Deaths from natural causes (old age, disease -- always active).
BirthsT1 / BirthsT2	New offspring this day.
AvgConditionT1 / AvgConditionT2	Average health of the population (0 = dead, 1 = perfect health).

4b. Aggregate CSV (one per run)

Summarizes all scenarios within one run. Sections are separated by === TITLE === headers.

Summary section:

Field	Description
Scenarios Run	Total number of scenarios executed.
Survived / Crashed	How many scenarios survived vs had total population collapse.
Crash Rate	Percentage of scenarios that crashed.
Avg Crash Day	Average day when crashes occurred (only shown if any crashed).

Per-Species Population Stats (the key section for Brian's request):

Every species gets its own row with statistics across all scenarios. Custom species are tracked individually by name (e.g. "Coral_Custom", "Nudibranch_Custom").

Column	Description
Species	Species name and variant (e.g. Hexapod_Arctic, Coral_Custom).
Avg	Average final population across ALL scenarios (including extinct ones showing 0).
SurvivedAvg	Average final population only across scenarios where this species survived. More meaningful -- shows what it achieves when alive, not diluted by extinctions.
Min	Smallest final population across any scenario.
Max	Largest final population across any scenario.
Extinct	Number of scenarios where this species went extinct (final population = 0).
Survived	Number of scenarios where this species survived (final population > 0).
ExtinctionRate	Percentage of scenarios where this species went extinct.

Individual Scenarios table:

One row per scenario showing seed, crash status, final populations by variant, and temperature stats. Includes T1Custom/T2Custom columns.

4c. Bulk Summary CSV (one per bulk upload)

Found at the ZIP root as bulk_summary.csv. Aggregates results across ALL runs in a bulk upload.

Per-Run Results table:

One row per CSV row (run), showing: run name, scenarios survived/crashed, crash rate, base temperature, climate trend, and average population for each species.

Per-Species Aggregate (across all runs):

Column	Description
Species	Species name_variant (same format as aggregate.csv).
GrandMean	Average of each run's average population. Note: if runs have very different temperatures, this mixes unlike conditions.
SurvivedMean	Average population only across runs where the species survived. More meaningful than GrandMean when many runs cause extinction.
RunsExtinct	Number of runs where this species went extinct.
RunsSurvived	Number of runs where this species had survivors.
ExtinctionRate	Percentage of runs where this species went extinct.

Note: There are no Min/Max columns at the bulk level. Min/Max of averages are not meaningful real population values -- use the per-run aggregate.csv files for Min/Max.

5. Default Species Quick Reference

Default thermal parameters for the three standard variants. These are the starting points when you select a variant in the Species Config screen.

Parameter	Arctic	Common	Tropical
Optimal Temp	18 C	24 C	30 C
Arrhen Breadth	4,000 (narrow)	8,000 (wide)	4,000 (narrow)
Arrhen Lower	13,974	3,000	15,827
Arrhen Upper	35,000	35,000	35,000
Lower Bound	17.75 C	23 C	29.75 C
Upper Bound	17.95 C	25 C	29.95 C
Peak Height (Pmax)	0.85 (specialist)	0.65 (generalist)	0.85 (specialist)

Key insight: Arctic and Tropical are specialists -- high peak (0.85) but very narrow thermal windows. Common is a generalist -- lower peak (0.65) but much wider tolerance. This means Common dominates at moderate temperatures, while specialists only thrive when conditions closely match their optimal range.

Default Biology Parameters

Parameter	Prey (Tier 1)	Predator (Tier 2)	Why Different
Starting Pop	1,000	100	Prey outnumber predators ~10:1
Eating Amount	0	1.5	Prey don't eat other species
Repro Multiplier	0.45	0.1	Prey reproduce ~4.5x faster
Death Threshold	0.3	0.3	Same stress tolerance
Death Rate	0.6	0.3	Prey die faster from stress (smaller = less buffering)
Natural Death Rate	2%	1%	Prey have shorter lifespans
Hunting Efficiency	N/A	75%	Only predators hunt

Auto-generated reference. For the latest details, check the simulation source code or contact the team.